2024 IEEE 9th International Conference for Convergence in Technology (I2CT)
Pune, India. Apr 5-7, 2024

Development of an E-Voting Scheme through a Blockchain-Enabled Approach with Deep Hybrid Learning for Biometric Authentication

Md Abdul Based
Department of Computer Science & Engineering
Dhaka International University
Dhaka, Bangladesh
Email: baset123@gmail.com

Rownak Borhan
Department of Computer Science & Engineering
Southeast University,
Dhaka, Bangladesh
Email: rownakborhan34@gmail.com

Khandaker Mohammad Mohi Uddin
Department of Computer Science & Engineering
Dhaka International University
Dhaka, Bangladesh
Email: jilanicsejnu@gmail.com

Mohammad Shorif Uddin
Department of Computer Science & Engineering
Jahangirnagar University
Dhaka, Bangladesh
Email: shorifuddin@juniv.edu

2024 IEEE 9th International Conference for Convergence in Technology (I2CT) | 979-8-3503-9447-4/24/$31.00 ©2024 IEEE | DOI: 10.1109/I2CT61223.2024.10543350

Abstract— **Developing a secure and transparent electronic voting (e-voting) scheme that maintains the fairness and privacy of traditional voting methods has long been a big challenge. In this research, a novel approach is presented that leverages deep hybrid learning techniques to enhance the biometric authentication system, combining various machine learning models such as Support Vector Machine, Random Forest, and K-Nearest Neighbors with deep learning architectures including VGG16, VGG19, Xception, and Inception. After authentication it extends to the design of a blockchain-based voting scheme, powered by smart contracts and NEAR wallet integration. This innovative model constructs a robust and secure blockchain-based e-voting solution. This research represents a significant step towards achieving a balance between the transparency and flexibility offered by electronic voting schemes while upholding the integrity and privacy crucial to the democratic process.**

Index Terms*— *E-voting, Machine Learning, Deep Learning, Blockchain, Fingerprint

I. INTRODUCTION

The process of voting that utilises electronic instruments, either partially or exclusively, is commonly referred to as electronic voting or e-voting. The objective of e-voting is to offer a range of sophisticated attributes. In essence, an e-voting scheme ought to guarantee both privacy and verifiability. The initial measure serves to impede the retrieval of an individual's (voter's) vote, while the subsequent measure enables each voter to confirm the presence of his/her vote on the public display and guarantees that the final tally aligns with the votes cast by the eligible voter. Furthermore, the validity of the scheme is contingent upon the inclusion of votes that have been generated in a fair and honest manner when determining the election's outcome. In addition to other desirable attributes for electronic voting schemes, there are robust types of privacy including receipt-freeness, coercion-resistance, and ballot independence.

The establishment of security features for e-voting schemes has been a subject of ongoing discussion, with the development of secure e-voting protocols that encompass these properties proving to be more complex compared to traditional paper-based voting systems. Numerous suggestions have emerged in recent years, exhibiting various categorizations based on the methods employed to ensure privacy and accomplish the tally function. Nevertheless, it is worth noting that the security of any protocols that can be proven to be secure continues to depend on classical assumptions [1]. This implies that the aforementioned techniques may be vulnerable to compromise in the event of the emergence of effective quantum computers. Hence, the task of developing a quantum-resistant electronic voting scheme presents significant difficulties, yet holds potential for instilling confidence among individuals in adopting e-voting protocols.

The proliferation of e-voting is impeded by the intricate nature of electronic systems and processes. The lack of transparency in numerous electronic voting schemes can result in a lack of information for both voters and election officials. The preservation of the election process's integrity mostly hinges upon a limited number of system operators as opposed to a multitude of poll workers, while the comprehension of most e-voting schemes is limited to a selected group of experts.

The deployment of e-voting schemes, if executed poorly, has the potential to undermine public confidence in electoral processes. This necessitates a thorough consideration of its execution and a comprehensive analysis of the historical context of e-voting.

The objective of this study is to address the security concerns associated with the blockchain technology used in the e-voting scheme, while also enhancing the biometric authentication system through the implementation of a deep hybrid learning approach.

II. RELATED WORKS

A concept for a multimodal verification system that incorporated the features of the face and iris at the feature level was developed by Bouzouina et al. [2]. The research made use of a variety of techniques to extract features and an algorithm called Support Vector Machine (SVM) to evaluate the users.

Ratha et al. [3] developed a technique of fingerprint authentication that is based on a graph representation and can handle fingerprints that have been altered in just one way. A weighted graph of fingerprint minutiae is created using the features of the fingerprint minutiae, and it is created for both the query fingerprint and the reference fingerprint.

The biometric identity system developed by Veluchamy and Karlmarx[4] makes use of uncommon physical characteristics, such as finger veins and knuckles, to create a system that may be applied in a variety of contexts. At the

Authorized licensed use limited to: Southeast University. Downloaded on October 18,2026 at 04:53:43 UTC from IEEE Xplore. Restrictions apply.

level of the features, the system merged the characteristics. The system had a good accuracy rate, and they achieved this through the application of the K-SVM algorithm.

Chanukya et al. [5] developed a multimodal biometric verification system with the help of a neural network. This approach achieved a success rate of 96.6%.In [6], in order to extract features for iris recognition systems, the Gabor filter and the 2-D wavelet filter are utilized. This method is unaffected by being moved or twisted, and it is also indifferent to the direction in which it is illuminated. According to the database maintained by the Institute of Automation of the Chinese Academy of Sciences (CASIA), the Gabor has an accuracy of 98.3%, whereas the wavelet has a rate of classification of just 82.5%.

In the procedure that is described in [7], multichannel and Gabor filters are utilized in order to obtain local information regarding the texture of the iris. After that, a feature vector with a fixed length is constructed using this information. The findings from the CASIA database indicated that the False Acceptance Rate (FAR) was 0.01%, while the False Rejection Rate (FRR) was 2.17%.

In addition, Amour et al. [8] extracted features for a multimodal biometric system that made use of characteristics of the iris and the face. The features of the iris were extracted with the use of a multi-resolution 2D Log-Gabor filter. On the other hand, the facial features were extracted with the assistance of singular spectrum analysis and normal inverse Gaussian.

Aguilar et al. [9] suggested a way to improve fingerprint imaging by combining Fast Fourier Transform (FFT) and Gabor filters. This combination was used in their method. After that, a one-of-a-kind step for recognition that takes into account regional features and statistical criteria is put into action. The solution that has been presented makes use of both thumbprints. The processing of each fingerprint is done on its own, and then the results of the many unimodal analyses are compared one by one in order to provide the final merged result. Experiments have been run using a fingerprint database that contains the prints of fifty persons and has a false alarm rate of 0.2% and a false positive rate of 1.4%.

A multimodal CNN was proposed by Soleymani et al. [10], which would incorporate iris, facial, and fingerprint features at various layers of the CNN. In order to integrate information, the authors employed a number of different techniques, including multi-abstract fusion, weighted feature fusion, and so on. According to the findings of the analysis, the optimal outcome was achieved by combining all three forms of biometrics.

It is abundantly evident, based on what has been mentioned previously that the machine learning algorithm known as SVM [2, 11, and 12] is the one that has been utilized the majority of the time in recent studies. Additionally, machine learning was applied to the process of recognizing the many types of biometrics. Using the deep learning methodology, the network extracts the features from the images in an autonomous fashion. There are times when the images require going through numerous stages of pre-processing [2, 11, and 12]. The performance of approaches based on deep learning is typically superior to that of methods based on machine learning in the majority of cases. Deep hybrid learning has not been used much in recent studies, despite the fact that it has the potential to be quite effective.

For governments and institutions, Agora [13] is a blockchain-based end-to-end verified voting solution. For elections, Agora employs its own Token on the blockchain, which governments and institutions buy for each qualified voter. Boardroom Smart Contract Voting with Maximum Voter Privacy [14] suggested the first implementation of the Open Vote Network (OVN), a decentralized and self-tallying internet voting protocol. The public Ethereum blockchain hosts the OVN as a smart contract.

Blockchain technology was proposed to be included into the UK's voting system [15], which allows voters to cast their ballots in person at a polling place or remotely via a web browser. On the Ethereum blockchain, there is a decentralized voting network called Netvote [15]. Decentralized applications (dApps) are used by Netvote for the system's user interface. Election administrators can design ballots, set registration guidelines, create ballots, and open and shut voting using the admindApp.

Though the existing publications have focus on security system-based blockchain technology, there was no deep hybrid learning-based biometric system for authentication of the voters. The blockchain-based voting database, and machine learning and deep learning techniques are used in this proposed electronic voting scheme. The scheme enriches the authentication process using machine learning and deep learning; and then uses blockchain technology as a database, to cast, create, and count ballots. Then, the scheme publishes the results accordingly.

III. METHODOLOGY

In this proposed e-voting scheme, a biometric authentication system has been developed using deep hybrid learning which is a combination of Deep Learning (DL) and Machine Learning (ML). Mainly biometric identification employing fingerprint and iris isemployed in the scheme. The SOCOFing [16] dataset is used for fingerprint and the MMU [17] dataset is used for iris. Total 6000 fingerprint images and 460 iris images are applied in the experiments. Then a blockchain-based voting scheme is developed using NEAR wallet to handle the transaction. All the e-voting transactions are done on a smart contract deployed in the blockchain.In this e-voting scheme, smart contracts generate a unique key(which is generated once)for each and every vote for the verification purpose. After the voting period is over, the final result is published.

A. Improved Authentication System

There is a great debate surrounding the use of biometric technologies, which play an important role in many scientific contexts. A novel biometric identification system using fingerprint and iris is proposed in this work. Multimodal biometric identification system is chosen since it takes into account more than one identifying characteristic, either physical or behavioral, to decrease the system's False Acceptance Rate (FAR) and increase its False Rejection Rate (FRR). In this study, the proposed machine learning approach is employed using deep learning to merge fingerprint and iris. By utilizing Gabor filters and Convolutional Neural Networks (CNN), the feature vector is generated. In order to improve the precision of fingerprint and irisclassification results, a deep learning algorithm and data augmentation were applied

Authorized licensed use limited to: Southeast University. Downloaded on October 18,2026 at 04:53:43 UTC from IEEE Xplore. Restrictions apply.

to the feature vectors. Three separate contexts are considered in which fingerprint and irisscore are used for recognition. Multiple score methods, including early fusion, late fusion, and intermediate fusion were used.

In this research, a deep learning-based ensemble model is presented. In biometrics, the two most frequent fusion methods are early fusion and late fusion.

In this proposed model, a deep learning framework and a fusion of hybrid machine learning algorithms are used to improve performance on fingerprint and iris datasets. The Gabor filter is utilized for both preliminary processing and final augmentation. An extractor is included in the DL algorithm. The images were accurately classified following the application of the classification method. The proposed model's final accuracy was achieved when an operation was applied to a combination of fingerprint and iris scans.

The True Positive Rate (TPR) and False Acceptance Rate (FAR) of the proposed model are calculated. The number of biometric samples (s) successfully identified is then divided by the total number of samples to get the TPR. The mathematical formula for calculating TRP is shown in Equation 1.

$$TPR = \frac{s}{No.of\ samples} \times 100 \quad (1)$$

The unit of measurement is a percentage (%). The FAR, which measures the average proportion of incorrect acceptances to the total number of biometric samples, is used in biometric recognition.Equation 2 helps to calculate FAR.

$$FAR = \frac{Prob.of\ incorrect\ recognition\ of\ Samples}{No.of\ Samples} \times 100 \quad (2)$$

Through the use of common real-time public databases, experiments confirmed that the proposed solution which employs a combination of machine learning techniques like the Random Forest, Support Vector Machine (SVM), Logistic Regression, KNN, and deep learning algorithms like VGG16,VGG19, Inception, and Xception outperforms theother approaches.

B. Blockchain-based E-Voting Scheme

The block diagram of the proposed voting scheme is shownin “Fig. 1”.In this scheme, first of all a smart contract is deployed in blockchain. All the transactions like poll creation, generation of keys, and vote counting are done in this smart contract.

In the scheme of “Fig. 1”, only the authenticated voters (authentication is done using biometric identification which is a combination of fingerprint and iris) will get access of the voting protocol. After authentication, a voter will choose a candidate and will cast the vote accordingly. During this time of voting, a unique transaction ID and a unique key will be generated in the smart contract. The unique key is anonymous, that is not identifiable by the external stakeholders. A voter will later be verified by this unique key. It will also ensure the privacy of the voter while maintaining the eligibility/validity of the voter since only after successful biometric authentication the voter will get access of the voting protocol and then the unique anonymous key will be generated. This process is implemented in a public blockchain. As blockchain is immutable, for every operation a

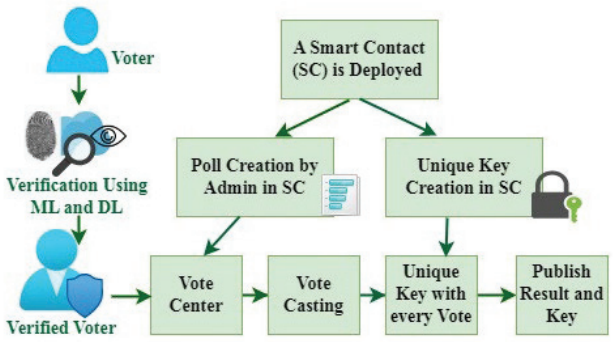


Fig. 1. Block Diagram of the Proposed Voting Scheme

unique key will be generated and it is not possible to delete any record from the smart contract. In order to deploy the smart contracts in a blockchain, a little amount of GAS price was paid.The followingEquation helps to calculate the gas price.

$$Gas\ Price = Gas\ limit \times Gas\ Price\ per\ Unit \quad (3)$$

In this operation for every node mining a .0.25 NEAR gas price was counted. For every block mining smart contract provides a hash value. Here voters are allowed to cast a vote for a limited period of time and a voter is allowed to vote only once. If a voter tries to cast a vote twice then the button to cast the vote willnot work. After the voting period is over, the resultwill appear on the website. In the result, the total vote will be counted for each candidate and for every voter the unique key will appear as a transaction token. This will allow every voter to verify that his/her vote is counted properly.

IV. EXPERIMENTS AND RESULTS

Experiments were conducted on a local PC utilising Google Colab to evaluate the efficacy of an enhanced authentication system, adhering to the proposedmethods. This study examines and evaluates the deep learning capabilities of many Python libraries, including NumPy (version 1.0.1), Pandas (version 1.11.0),Scikit-learn (version 1.0.1), Seaborn (version 1.0.1), and Matplotlib (version 1.0.1).

The inquiry heavily employs the utilisation of deep hybrid learning. In the internally curated fingerprint and iris databases, the K-Nearest Neighbors (KNN) algorithm demonstrated superior performance compared to other machine-learning classifiers, with an accuracy rate of 92%.

The accuracy of the system is improved through the utilisation of a mixed machine learning approach. In this case, VGG16, VGG19, ResNet-50,Xception, and Inception are used. The Xception method effectively mitigates several deep learning algorithms, resulting in an observed accuracy rate of 82%. The VGG-16 model achieves an accuracy of 80%, while the VGG-19 model achieves an accuracy of 79% and Inception achieves an accuracy of 73%. On the other hand, the ResNet-50 model achieves an accuracy of 65%. In this experiment, Xception has demonstrated the highest level of accuracy (82%). Additionally, it is worth noting that there are a total of 50,324 trainable parameters associated with this model.

In the conducted experiment, in order to enhance the precision of the results, a deep hybrid learning approach is employed. The approach first involved conducting

Authorized licensed use limited to: Southeast University. Downloaded on October 18,2026 at 04:53:43 UTC from IEEE Xplore. Restrictions apply.

preprocessing. Image preprocessing involves the use of morphological operations, such as zooming, shifting, left shifting, and right shifting. The experiment employed the utilisation of Gabor filters and the application of transfer learning techniques.

Various deep learning algorithms are employed in the transfer learning process, including VGG16, VGG19, Inception, and Xception. Gabor filters and deep learning algorithms are employed to extract features from images. Two features are encoded in a single unit vector. The aforementioned method is conducted for both fingerprint data and iris data. The suitability of this unit vector for a machine learning classifier is influenced by certain factors. Multiple classifiers such as Random Forest, Logistic Regression, SVM, and Decision Tree have demonstrated superior performance when applied in conjunction with transfer deep learning techniques in comparison to alternative methods. After doing classification, a logical AND operation was performed on the two resulting outputs in order to determine the optimal algorithm for deep hybrid learning.

The accuracy of the DL algorithms and deep hybrid learning algorithms are shown in "Fig. 2" and "Fig. 3" respectively. As shown in "Fig. 2", Xception has demonstrated the highest level of accuracy. For classification, Random Forest (RF) provides the highest accuracy among other machine learning classifiers. As shown in "Fig. 3" , Xception+RFprovides the best accuracy compared to others.

The comparison between different systems in the literature and the proposed system in this paper is shown in Table I.

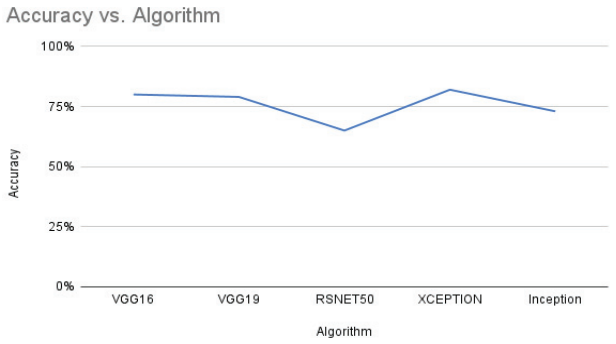


Fig. 2. Accuracy of DL Algorithms

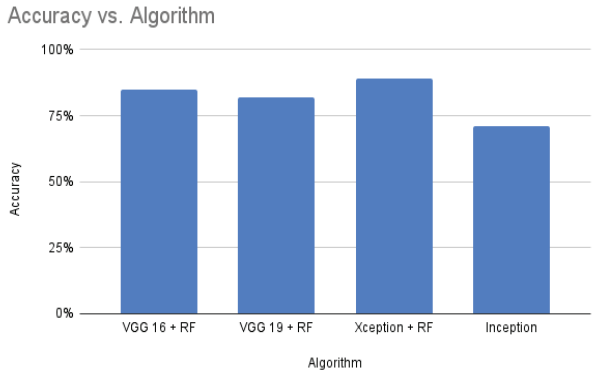


Fig. 3. Accuracy of Deep Hybrid Learning Algorithms

TABLE I. COMPARISON BETWEEN THE EXISTING SYSTEMS AND THE PROPOSED SYSTEM.

Reference	Technique	Dataset	Accuracy
Bouzouina et al. [2]	SVM	Multimodal dataset(Face and Iris)	98.8%
Chanukya et al. [5]	Modified region growing algorithm and gabol filter	Fingerprint and Ear image	96.6%
Veluchamy and Karlmarx et al. [4]	K-SVM	Finger veins and Knuckles	96%
Z. Zhao et al.[6]	Gabor filter and the 2-D wavelet filter are used to pull out	Iris	82.5%
Proposed System	Hybrid approach of ML and DL	Fingerprint and Iris	94.5%

Though Table I shows a lower level of accuracy of the proposed system, it is important to note that the deep hybrid learning approach is implemented in the proposed system. In comparison to previous approaches in deep hybrid learning techniques, this method achieved the highest level of accuracy. In addition, the utilisation of fingerprint and iris data classifications results in a reduced false rate, as a late fusion technique was implemented. The present system processes all images acquired by a sensor, resulting in instances when data may be flawed and the accuracy rate is diminished. In previous studies, researchers utilised an optimal dataset to evaluate the efficacy of their methodology. The rationale for employing deep hybrid learning is in its ability to effectively perform feature extraction through deep learning. Currently, this particular approach was not found in any published literature.

Then in blockchain portion of this proposed e-voting scheme, first of all a voter needs to be authenticated by using biometric identification (fingerprint and iris). Then the voter is redirected to the voting web page. Accordingly the voter chooses the candidate and casts vote respectively. The voter needs to be logged in using his/her NEAR wallet account. In testnet voters do not need any NEAR coin but in MAINNET voters need 0.25NEAR coin for every vote as GAS price.After casting a vote, the voter will be removed from the website and it is noted that voters cannot do the voting action twice. After voting period is over, the admin will publish the result page.

In the admin section, first of all they need to deploy the smart contract in the blockchain. In blockchain, smart contracts are deployed by NEAR dev. "Fig. 4" depicts every vote and count along with the encrypted unique keys which can be used for verification purpose. The unique keys allow the voters to check iftheir votes have been counted properly. The keys are unique and encrypted, hence does not reveal the identity of any voter.

Authorized licensed use limited to: Southeast University. Downloaded on October 18,2026 at 04:53:43 UTC from IEEE Xplore. Restrictions apply.

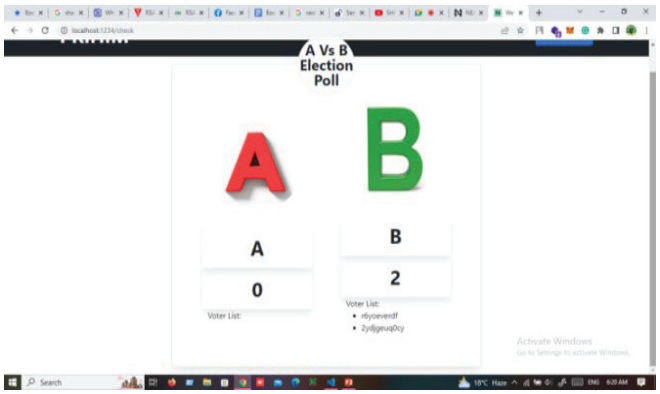


Fig. 4. Final Result

The smart contract is shown in “Fig. 5” with a unique transaction ID. A snapshot of the transactions in blockchain is shown in “Fig. 6”. “Fig. 7” shows a voter’s transaction ID paneland “Fig. 8” shows the details of a transaction in blockchain that is used in the proposed voting scheme.

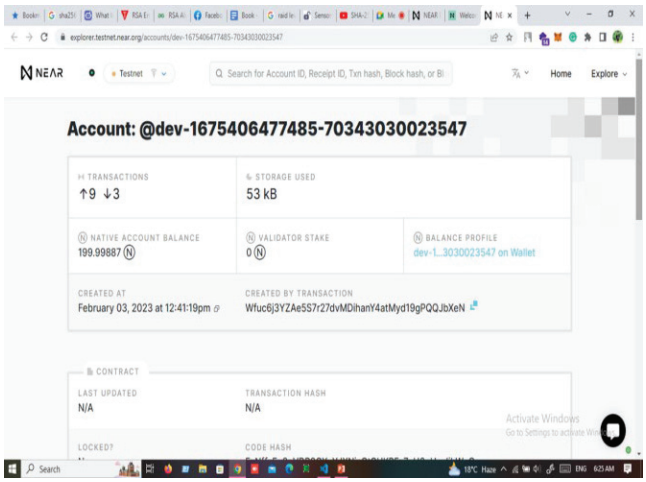


Fig. 5. Smart Contract Page

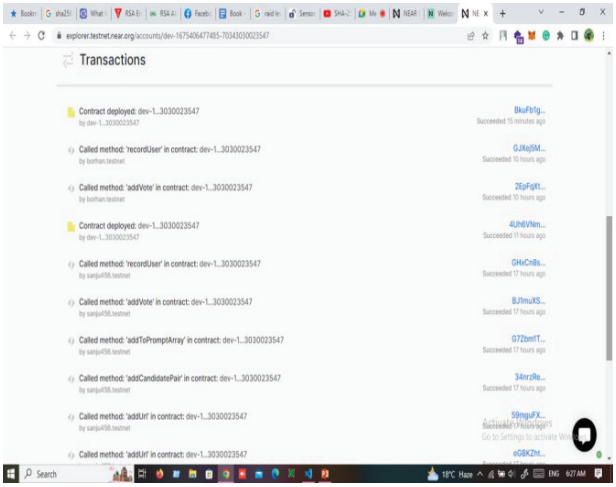


Fig. 6. Snapshot of the Transactions in Blockchain

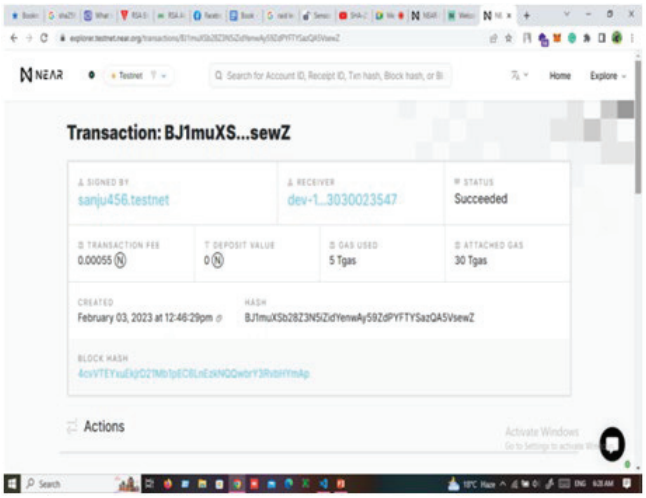


Fig. 7. A Voters’Transaction ID Panel

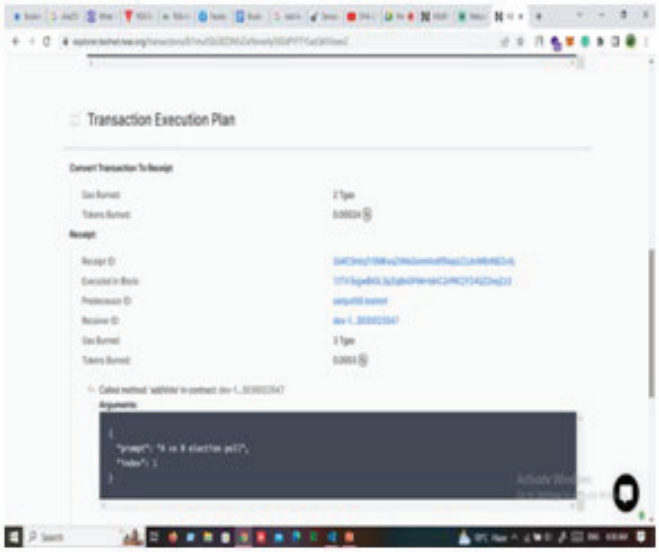


Fig. 8. Details of One Transaction in Blockchain

V. CONCLUSIONS

Blockchain-based e-voting after authentication employing fingerprint and iris is a concept that leverages the decentralized and secure nature of blockchain technology to create an electronic voting scheme that can be used for various types of elections. Improved accessibility for impaired voters is another potential benefit of electronic voting technology, which also aims to speed up the counting of ballots and cut down on the cost of hiring people to count votes manually. It takes less time to report and publicize findings. In this paper, voter authentication is done using biometric identification (fingerprint and iris). The authentication process is a novel one that leverages deep hybrid learning techniques combining various machine learning models such as SVM, Random Forest, and K-Nearest Neighbors with deep learning architectures including VGG16, VGG19, Xception, and Inception.

In the proposed scheme, votes are recorded on a blockchain and the voting process is transparent, secure, and tamper-resistant. By using the encryption and decentralized architecture, blockchain-based e-voting can reduce the risk of manipulation of the voting process. The use of the unique key ensures that every vote is recorded correctly and can be audited, thus increasing transparency and accountability in the

Authorized licensed use limited to: Southeast University. Downloaded on October 18,2026 at 04:53:43 UTC from IEEE Xplore. Restrictions apply.

voting process. This blockchain-based e-voting scheme has the potential to improve the security, transparency, and accessibility of the voting process though further analysis with multiple datasets and testing are needed to determine its feasibility and effectiveness.

REFERENCES

[1] T. Kohno, A. Stubblefield, A. D. Rubin, and D.S. Wallach,"Analysis of an electronic voting system", In IEEE Symposium on Security and Privacy, 2004. Proceedings. 2004, pp. 27-40

[2] Y. Bouzouina, and L.Hamami, "Multimodal Biometric: Iris and face recognition based on feature selection of Iris with GA and scores level fusion with SVM", In proceedings of 2017 2nd International Conference on Bio-Engineering for Smart Technologies, Paris, France, 30 August-1 September 2017, pp.1-7

[3] N. K. Ratha, R. M. Bolle, V. D. Pandit, and V. Vaish, "Robust fingerprint authentication using local structural similarity", in Proc. 5th IEEE Workshop Appl. Comput. Vis., Dec. 4–6, 2000, pp. 29–34. DOI 10.1109/WACV.2000.895399.

[4] S.Veluchamy; and L.R.Karlmar,"A System for multimodal biometric recognition based on finger knuckle and finger vein using feature-level fusion and k-support vector machine classifier",IET Biometrics. 2017, 5. 232-242.

[5] P.S.V.V.N.Chanukya,and T.K. Thivakaran."Multimodal biometric cryptosystem for human authentication using fingerprint and ear". Multimed Tools Appl 79, 659–673 (2020). https://doi.org/10.1007/s11042-019-08123-w.

[6] Z. Zhao, and K. Ajay. "An accurate Iris segmentation framework under relaxed imaging constraints using total variation model," in Proceedings of the IEEE International Conference on Computer Vision, 2015, pp. 3828–3836

[7] A.S. Al-Waisy, R.Qahwaji, S.Ipson, S. Ai-Fahdawi, and T.A.M. Nagem, "A multimodal-biometric Iris recognition system based on deep learning approach", Pattern Anal. Appl. 2018, 21, 783-802

[8] N. Ammour, Y. Bazi, and N.Alajlan, N. "Multimodal Approach for Enhancing Biometric Authentication", J. Imaging 2023, 9, 168. https://doi.org/10.3390/jimaging9090168

[9] G. Aguilar, G. Sanchez, K. Toscano, M. Nakano, and H. Perez, "Multimodal biometric system using fingerprint," in Proc. Int. Conf. Intell. Adv. Syst. 2007, pp. 145–150. DOI: 10.1109/ICIAS.2007.4658364.

[10] S. Soleymani, A.Torfi, J. Dawson, and N.M. Nasrabadi, "Multi-Level Features Abstraction from Deep Convolutional Neural Networks for Multimodal Biometric Identification". In Proceedings of the 2018 24th International Conference on Pattern Recognition, Beijing, China, 20-24 August 2018; pp. 3469- 3476

[11] A. Navada, A. N. Ansari, S. Patil and B. A. Sonkamble "Overview of use of decision tree algorithms in machine learning." 2011 IEEE control and system graduate research colloquium. IEEE, 2011.

[12] G. Guo, H. Wang, D. Bell, Y Bi, and K. Greer. "KNN model-based approach in classification." OTM Confederated International Conferences"On the Move to Meaningful Internet Systems". Springer, Berlin, Heidelberg, 2003.

[13] Agora (2017). Agora: Bringing our voting systems into the 21st century.Available at: https://agora.vote/Agora_Whitepaper_v0.1.pdf

[14] P. McCorry, S. F. Shahandashti, and F. Hao,"A Smart Contract for Boardroom Voting with Maximum Voter Privacy."In book: Financial Cryptography and Data Security, 2017, Available at: https://eprint.iacr.org/2017/110.pdf

[15] A. Jonathan, L. Steven, and H. Ben,"Netvote: A Decentralized Voting Network", 2018, Available at: https://netvote. io/wp-content/uploads/2018/02/Netvote-White-Paper-v7.pdf

[16] Sokoto Coventry Fingerprint Dataset. Kaggle. Date: 22/06/2023https://www.kaggle.com/ ruizgara/socofing

[17] Multimedia University Dataset. Kaggle. Date: 20/05/2023https://www.kaggle.com/datasets/naureenmohammad/mmu-iris-dataset

Authorized licensed use limited to: Southeast University. Downloaded on October 18,2026 at 04:53:43 UTC from IEEE Xplore. Restrictions apply.